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CMPT417 - Hacking and Pen Testing
Website Password

This assignment utilizes a public website made for educational purposes. It investigates the use of inspecting a webpage's source code to find the username and password to successfully log in at differing levels of difficulty.

The website listed below is a public website used for education and training purposes that lets you test your hacking skills; it's safe and legal to try hacking this site (ignore any advertising or buttons telling you to download stuff, just try to work out the credentials to get past each level). Using the techniques discussed in class, see if you can get past Level 1 and Level 2. Submit the correct password on iLearn along with a short description of how you did it, with enough detail that someone else could reproduce what you did from your description.

https://www.try2hack.lt/en/

There are 10 different levels on this site, each progressively more difficult. We don't have time in class to cover all the ways you can hack into each level. But you may choose to try this on your own time. Email me the correct login credentials for each additional level and your description of how you got them (in your own words, with enough detail that someone else could reproduce what you did) to earn +1 point of extra credit per level.

LEVEL 1

By inspecting the source code of the login field, the password can be found within the script that checks for password validation.

Password: l4m3rz

```
<script language="JavaScript">
      function Try(slp) {
        if (slp == "l4m3rz") {
          alert("Great! Go to the 2nd level...");
          location.href = "level2-room.htm";
        }
        else {
            alert("Wrong password. Go away...");
            location.href = "about:blank";
        }
      }
    </script>
```

LEVEL 2

In this level, the mouse right-click function is turned off. Luckily, there is another way to navigate to the page's source code. On Google Chrome, navigate to the menu bar, click the "More tools" option, and select "Developer tools." The keyboard shortcut for this is Ctrl+Shift+I to open the inspect element panel. Once the inspect panel is open, mouse over the section of the page containing the username and password textboxes and open the script tags until the one containing the user and password is found.

```
▼<script language="JavaScript"> == $0
    function Tryi(uzer,pswd){if(uzer=="ozzy" && pswd=="osburn"){alert("Great! Go to the 3d
    level...");location.href="level3-kbbe.htm";}else{alert("Wrong username and/or password. Go
    away...");location.href="about:blank";}}
  </script>
```

Looking more closely at the code, when the function Tryi is called, the password is passed to the function as the username and the username is passed to the function as the password. Essentially, the username and password are swapped.

```
<input type="reset" value="Click if you want to continue" onclick="Tryi(testi.pass.value,
testi.uzr.value)">
```

Username: osburn **Password:** ozzy

LEVEL 3

This level requires the user to enter the password into a popup, so the page's source code cannot be accessed like the first two levels. Instead, copy and paste the link to level 3 found in the source code for level 2 ("`level3-kbbe.htm`") into the end of the URL for the level 2 source code. Now the source code for level 3 can be accessed.

New URL:

```
view-source:https://www.try2hack.lt/en/lygiai/level3-kbbe.htm
```

```
<script language="JavaScript">
  slp = prompt("Please enter the 3d level password:","");
  if (slp == "fatman_")
  {
    alert("Great! Go to the 4th level...");
    location.href="level4-blok.htm";
  }
  else
  {
    alert("Wrong password. Go away...");
    location.href="about:blank"
  }
```

Password: fatman_

LEVEL 4

This level uses flash to accept the password, so it is a bit more challenging especially now that flash is no longer supported. There is also a fake password “bullet” within the source code that does not actually work. Instead, the embed link holds the file “level4.swf” which needs to be decompiled to see the source code.

Fake Password:

```
<script language="FlashMX">
      function Key(pass) {
        if (pass == "bullet") {
          alert("Great! Go to the 5th level...");
        }
        else {
            alert("Wrong password. Go away...");
        }
      }
    </script>
```

Embed Link:

```
<embed src="level4.swf" quality="high" bgcolor="#ffffff" width="300" height="90" type="application/x-shockwave-flash"
```

Downloading the level4.swf file (Note: if this were not a site built for educational purposes, it would be a bad idea to download random files on a personal device) and decompiling it with a free online flash decompiler:

https://pdfrecover.herokuapp.com/swfdecompiler/

results in a zipped file of the flash source code. The file DOMDocument.xml contains the real code for the password field.

Real Password:

```
▼<script>
   <![CDATA[ on(release, keyPress "<Enter>"){ if(_root.pass == "bullet") { getUrl("level5-utia.htm", "_self"); }
   else if(_root.pass == "EasySWF") { getUrl("level5-jkli.htm", "_self"); } else { _root.pass = ""; } } ]]>
 </script>
```

Again, using the URL associated with the fake password “bullet” brings you to the incorrect level 5 source code, but the real password “EasySWF” takes you to the next level. Because flash is no longer supported, the password cannot be entered. Instead, manually change the URL to access the level 5 source code again.

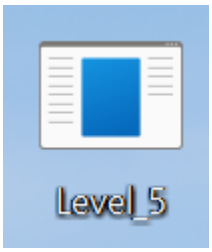
Password: EasySWF

Fake 5th Level Source Code:

```
<td colspan="4"><h1>This is not 5th level</h1><br>
```

LEVEL 5

This level also requires a file download. Within `Level_5.rar` is an executable called `Level_5.exe`.



Level 5 Download and Executable:

```
download the program and see the next level url from it.. Program is <a href="Level_5.rar">here
```

The application can no longer be run on the current version of Windows, but the code can be accessed in a text editor. Most of it is unreadable, but the username, password, and URL of the next level's source code can be found within.

```
www.try2hack.projektas.lt º▯º                          º▯º    ▯::[ Try to hack ]::▯    º▯ÇAAAAAAAAAAAAAAAAAAAAAAAAAAAA¶▯º Level 5th                º▯º Enter username and pass-
word, then you will see   º▯º url of the next level.    º▯º Type "exit" to exit.      º º      ▯º Username:                º▯º Password:
º▯È¿¿¿
```

LEVEL 6

This level has a fake username and password as well - the code containing the fake credentials is actually commented out. This URL again takes you to a false level 7 source code page.

Fake Credentials:

```
<script language="JavaScript">
  <!--
  uzr = prompt("LEVEL 6th. Please enter username:","");
  slp = prompt("LEVEL 6th. Please enter the password:","");
  if (uzr == username && slp == password)
  {
    alert("Great! Go to 7th level...");
    location.href=url;
  }
  else
  {
    alert("Wrong username and/or password. Go away...");
    location.href=burl;
  }
  username="Meatman";
  password="Psycho";
  url="level7-dkrh.htm";
  burl="about:blank";
  //-->
</script>
```

The only other script tag in this source code is found above this fake script, and provides a link to another file, "`src`". Opening this file reveals the correct credentials and URL for the next level's source code.

```
<script src="src"></script>
```

```
username = "LefTy"
password = "wIndow"
url = "level7-irra.htm"
burl = "about:blank"
```

Username: LefTy

Password: wIndow

LEVEL 7

Level 7 is similar to level 4 in that a flash file is provided and must be decompiled, but this level requires substring calculations in order to find the password and next source code URL.

```
<![CDATA[ _root.user = ""; _root.pass = ""; main = "ABCDEFGHIYJKLMNOPRSTUVZXQW0123456789"; vrt = main.substr(12,1) +
main.substr(1,1) + main.substr(12,1) + main.substr(15,0) + main.substr(20,1) + main.substr(21,1) + main.substr(19,1)
+ main.substr(1,1) + main.substr(23,2); ]]>
```

```
<![CDATA[ on(release, keyPress "<Enter>"){ a = _root.user; b = _root.pass; toliau = _root.main.substr(18,2) +
b.substr(3,1) + b.substr(6,1); if(a == vrt && b.substr(3,1) * b.substr(6,1) == 72) { getURL("level8-" + toliau +
".htm","_self"); } else { _root.pass = ""; _root.user = ""; } } ]]>
```

Given the string main, the username(vrt), the password (b), and the URL can be found. There are two possibilities for the password, but only one URL takes you to the real next level, which tells us which password is correct as well.

```
main = "ABCDEFGHIYJKLMNOPRSTUVZXQW0123456789";

vrt     = main.substr(12,1)     L
        + main.substr(1,1)      B
        + main.substr(12,1)     L
        + main.substr(15,0)     P
        + main.substr(20,1)     U
        + main.substr(21,1)     V
        + main.substr(19,1)     T
        + main.substr(1,1)      B
        + main.substr(23,2);    ZX

vrt = LBLPUVTBZX
```

```
toliau  = _root.main.substr(18,2)       RS
        + b.substr(3,1)                 8 or 9
        + b.substr(6,1);                9 or 8

getURL("level8-" + toliau + ".htm","_self");

url = "level8-RS98.htm"         b = XXX9XX8
            OR
      "level8-RS89.htm"         b = XXX8XX9
```

```
if(a == vrt && b.substr(3,1) * b.substr(6,1) == 72)

b = XXX8XX9 or XXX9XX8          8 x 9 = 72
    0123456    0123456
```

Username: LBLPUVTBZX ***Password***: XXX9XX8

URL: level8-RS98.htm

LEVEL 8

This level contains a downloadable .rar file which again contains an executable file that can be opened from a text editor

```
pass: Mole
user: eyeless level9-qwsa.htm gehog ckly html.htm ÉÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍ» º www.try2hack.projektas.lt º º
º ÇÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄÄ¶ º Level 8th                º º Enter username and pass-  º º word, then you will see
º Username:                 º º Password:                º ÈÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍÍ¼ exit Hed pri level9-
  WRONG DATAŠ  Ð š
```

This time, the information is less straightforward. There is a false username, password, and URL again, but by combining different sections of the fragmented data in the same pattern, the real credentials can be found.

Username: Hedgehog

Password: prickly

URL: level9-html.htm

LEVEL 9

Unlike the previous levels, the level 9 webpage uses PHP to process the login. Changing the webpage's extension of .htm to .php brings us to the PHP version of the level 9 webpage.

Use of PHP and New URL:

```
<form method="post" action="level9login.php">
```

try2hack.lt/en/lygiai/level9-html.php

Upon entering an invalid login, the URL changes. Using this, we can change the login field in the URL to set it equal to 1, reaching the webpage as if we had entered the correct password.

Incorrect Guess URL and Altered URL:

try2hack.lt/en/lygiai/level9-html.php?login=

try2hack.lt/en/lygiai/level9-html.php?login=1

A link now pops up on the webpage below the password field directing us to level 10.

LEVEL 10

I'm not sure if this was intentional or not, but upon leaving the user and password fields blank in level 10, it provides a link to level 11, which isn't finished yet.

Username: ***Password***:

LEVEL: 11th

Level 11th isn't done yet...

Please submit the form, and later you'll get an e-mail with message that 11th level is online. When the form is complete, you'll see all nicknames which already hacked all levels. Thank you for hacking our site :).

Nick:
e-mail:
Homepage: http://
Country: Lithuania
City:
Submit